Ahmedabad University

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School of Engineering and Applied Science

Brain Tumor Segmentation and Localization
Using Traditional Image Processing and OpenCV

Weekly Status Report
Week 5 of 10

Course: ECE501 – Digital Image Processing — **Instructor:** Professor Mehul Raval — **Section:** 01

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1 Introduction

In our project, classical image processing (no deep learning) is used for detecting and characterising brain tumors in the MRI scans. The dataset is the BraTS 2020 data set which provides four MR sequences per patient (T1, T1ce, T2 and FLAIR) along with an expert tumor outline for each MR scan. Each scan is cleaned, bright tumor regions are selected using thresholds, morphology operations are applied and the result is compared to the expert result, which is an outline of the tumors.

The first full version of the pipeline gave a mean Dice score of 0.665 for the Whole Tumor region on patients 001–020. It missed part of the tumor in many scans: it found only about 60% of the true tumor. This week we worked on that problem.

2 Objective

The goal this week was to improve the thresholding step of the segmentation so that it misses less of the tumor (less under-segmentation), without changing the rest of the pipeline. We also wanted the new threshold values to work on any patient, and not only on the 20 patients we used to choose them.

3 Literature Review

The BraTS challenge benchmark [1], its data descriptor [2] and the challenge summary [3] (the three papers to cite when using BraTS 2020) define the dataset, the four MRI sequences and the tumor labels we use. We follow the usual BraTS convention that the Whole Tumor region is the union of all tumor labels.

Intensity thresholding is a basic segmentation method. A traditional automatic method is Otsu's method (Otsu's 1979) that selects one threshold value from the image histogram. In fact, we use percentile thresholds: Voxels are retained if they are brighter than a desired percentage of all brain voxels. This is simple, but it always selects a fixed share of the brain, whatever the tumor size. Contrast-limited adaptive histogram equalization [5] is used in our preprocessing step to improve contrast.

4 Work Completed This Week

4.1 Methodology

Finding the problem. Our submitted rule keeps a voxel when it is bright in FLAIR and also bright in T2 or T1ce. The cut-offs were the 85th percentile for FLAIR, the 85th for T2 and the 90th for T1ce. This gave a precision of 0.79 but a recall of only 0.60, so the masks were too small. Small tumors did worst, because a fixed percentile then picks up bright normal tissue instead of the tumor.

What we tuned. We changed only the three percentile values. The preprocessing, the rule itself, the morphological steps and the 200 mm^3 minimum-size filter were left exactly as submitted. We tested 648 combinations: FLAIR and T2 from the 65th to the 97th percentile, and T1ce from the 70th to the 98th.

Avoiding over-fitting. With only 20 patients, it is easy to pick values that work well on these patients and nowhere else. To check this, we hid some patients from the tuning step:

- **Masked split:** 12 patients were used to choose the values, then the values were frozen and tested on 8 hidden patients.

- **Leave-one-patient-out:** for each patient, the values were chosen using the other 19 and tested on that one patient.
- **1000 random splits:** the same idea with 8 randomly hidden patients each time. This also shows how often each setting is chosen.

Settings were ranked by Dice plus half the recall, which gently favors less under-segmentation. Scores were also averaged with neighbouring settings, so that a lucky single value is not preferred over a stable region.

4.2 Implementation Details

We wrote a separate tuning package that does not modify the submitted pipeline or the data. We first rebuilt our baseline in this package and checked it: it gave a Dice of 0.2474 on patient 003, the same as our submitted pipeline. The morphological steps (opening and closing) were run on the GPU with PyTorch to make the 648-setting search fast, while hole filling and component labelling stayed on the CPU with SciPy. We checked that the mask from this faster path is identical to the submitted pipeline's mask. The full search over 20 patients took about 10 minutes. All per-patient scores, the chosen values and the reports are saved as files.

After the search, we confirmed the result with the real pipeline. We made a copy of the submitted code, changed only the three percentile numbers in `segmentation.py` (85/85/90 to 91/70/70), and ran `main.py` on patients 001–020, together with an unchanged copy. The unchanged copy gave a mean Dice of 0.6646, matching our earlier 0.665, and the changed copy gave 0.7240. These numbers equal those from the tuning package.

5 Results

The best setting was **FLAIR 91, T2 70, T1ce 70** (percentiles) instead of the submitted 85, 85, 90. Chosen in 20 of 20 leave-one-patient-out rounds, also in 55 % of random splits – thus it is not dependent on choice of patients.

Whole Tumor, BraTS 2020	**Submitted (85/85/90)**	**Tuned (91/70/70)**
Mean Dice, all 20 patients	0.664	0.724
Mean Dice, hidden patients (leave-one-out)	0.664	0.722
Mean Dice, hidden patients (1000 splits)	0.664	0.717
Recall (share of tumor found)	0.599	0.836
Precision	0.787	0.679
IoU (Jaccard), all 20 patients	0.521	0.583
HD95 boundary error in mm (lower is better)	28.1	39.4
Patients with recall below 0.70	14 of 20	4 of 20

Table 1: Submitted and tuned thresholds, measured with the real pipeline on patients 001–020. The hidden-patient rows use values chosen without those patients.

The Dice score on hidden patients (0.72) is close to the score on all patients (0.724), so we see no sign of over-fitting. "tuned" setting had a higher success rate than the submitted one in a leave-one-patient-out test, with a success rate of 14-20 patients in 90% of the random splits. Under-segmentation fell clearly: recall rose from 0.60 to 0.84. The cost is lower precision (0.79 to 0.68), meaning the masks are now slightly too large instead of too small. The boundary error (HD95) also became worse, from 28.1 to 39.4 mm on average, because the larger masks reach further from the true tumor edge in some scans. The spread across random splits was 0.64 to 0.78 Dice, so a small test group can still give a weaker result.

We also tried a different rule based on per-scan z-scores. It gave a higher Dice (0.80), but it changes the method itself, so we did not use it. We kept the original pipeline and changed only the thresholds.

Figure 1 shows one example, patient 010, which was one of the hidden patients (it was not used to choose the values). With the submitted thresholds the mask covers only the brightest core of the tumor (Dice 0.34).

With the tuned thresholds the mask also covers the weaker, swollen area around the core (Dice 0.65). Figure 2 shows the Dice and recall of every patient.

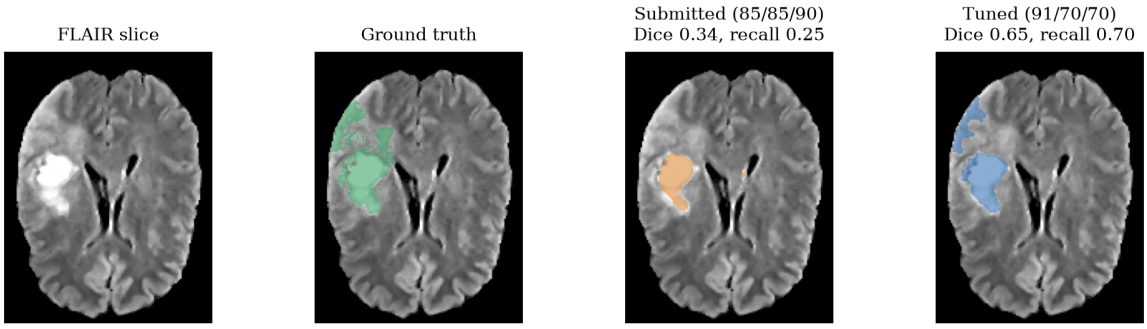


Figure 1: Patient 010 (axial slice with the largest tumor): FLAIR image, expert ground truth, and the masks from the submitted and the tuned thresholds.

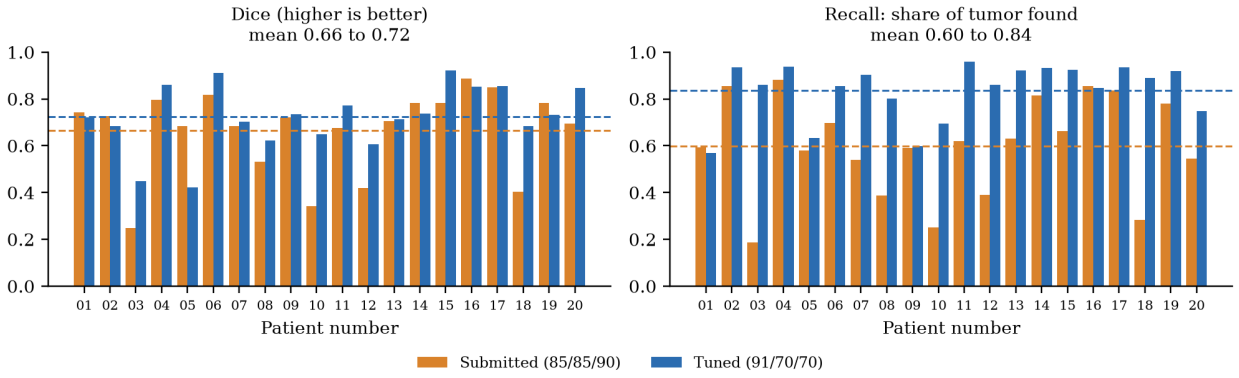


Figure 2: Dice and recall for each of the 20 patients, submitted versus tuned thresholds. Dashed lines show the mean.

6 Challenges Faced

- **Small dataset:** 20 patients are few for choosing values safely, which is why we used several kinds of hidden-patient tests.
- **Trade-off:** higher recall lowers precision. Pushing recall further (to 0.87) made masks about 70% too large and the Dice fell below the baseline on hidden patients, so we stopped at a balanced point.
- **Uneven gains:** the new values help some patients a lot (for example patient 010: Dice 0.34 to 0.65, and patient 003: 0.25 to 0.45). Five patients (001, 002, 014, 016, 019) drop by 0.02 to 0.05, and patient 005, and patient 005 drops clearly (0.68 to 0.42).

7 Plan for Next Week

- Place new percentiles (91, 70, 70) in the segmentation module of the repository, and update the result data in tables and figures for all 20 patients.
- Study the patients where the score dropped (especially patient 005), and the higher boundary error.
- Test the same values on more BraTS patients not used so far, to confirm that they generalize.
- Look at ways to recover precision, such as changes to the morphological steps, without losing recall.
- Begin the tumor localization part of the project.

References

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